

Discrete event traffic simulation

Part 2

Traffic simulation of packet-oriented loss and waiting systems

Completion date: 20th October

a) Loss system: Erlang-B distribution

Consider a packet-oriented loss system with N channels running at 1 Gbit/s each. Assume that the activity of all users who access the system is represented by a source that generates packets with an arrival rate $\lambda = 200$ packets/ms (Poisson process) and that each packet transmission has an average duration $d_m = 0.008$ ms, with exponential distribution (Erlang-B model).

Based on the event-driven simulation method, develop a simulation program of this system, enabling you to obtain, for a given N , the estimator of the blocking probability B (equal to the packet loss probability).

Compare the results of the blocking probability B with theoretical predictions (you may use, for example, the online traffic calculator available [here](#)).

b) Waiting system with infinite length queue: Erlang-C distribution

Adapt the previous program, considering that the packets that do not have available resources are placed in a queue with infinite length.

In this case, assuming that A is the packet delay, calculate:

- the estimator of the probability that a packet is delayed $P(A>0)$;
- the estimator of the average delay A_m of all packets;
- the histogram of the delay of the packets that suffer delay ($A>0$);
- the estimator of the probability that a packet is delayed more than A_x $P(A>A_x)$.

Compare the results of the **estimator of the probability that a packet is delayed** $P(A>0)$ and the **estimator of the average delay A_m of all packets** with theoretical predictions (you may use, for example, the online traffic calculator available [here](#)).

c) General case – waiting system with finite length queue

Adapt the previous program, considering that the packets that do not have available resources are placed in a queue with finite length L .

Show that in this case, the previous results of the Erlang-B and Erlang-C distributions can be obtained by making $L=0$ and L arbitrarily large, respectively. Compute the queue length L , which leads to a packet loss probability of 1%.

Compare your simulation results with theoretical predictions (you may use, for example, the online traffic calculator available [here](#)).